Question ID: c9417793

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$|x-9|+45=63$$

What is the sum of the solutions to the given equation?

Correct Answer: 18

Rationale

The correct answer is 18. Subtracting 45 from each side of the given equation yields $|x-9|=18$. By the definition of absolute value, if $|x-9|=18$, then $x-9=18$ or $x-9=-18$. Adding 9 to each side of the equation $x-9=18$ yields $x=27$. Adding 9 to each side of the equation $x-9=-18$ yields $x=-9$. Therefore, the solutions to the given equation are 27 and -9, and it follows that the sum of the solutions to the given equation is $27+(-9)$, or 18.